LAB 3

Objective:

The objective of this experiment is to understand the fundamental principles and techniques of digital post-mortem examination (digital autopsy) using the Autopsy open-source digital forensics platform. The experiment aims to provide practical knowledge of acquiring, examining, and analyzing digital evidence from storage.

Drive Link: (all files forensic, screenshot, recovered files, timeline, results are present here)
https://drive.google.com/drive/folders/1RBgSbabIInMUtixHyKaP_kaY8t10PgER?usp=sharing

1. Data Acquisition

I created a new partition Drive G on my laptop with a storage capacity of around 4 GB (3999 MB). The disk geometry showed a total sector count of 8,189,952 sectors, with 512 bytes per sector. I copied some of my project folders (frontend and backend files of one of my projects) , LLD system design videos, some screenshots from my laptop and text files. The total size of the drive was 1.2 GB.

I downloaded Autopsy and FTK Imager software from the internet and installed them. To acquire a forensic image safely without modifying the target drive, I opened FTK Imager and entered the case information as follows:

Case Number: CS445_LAB3

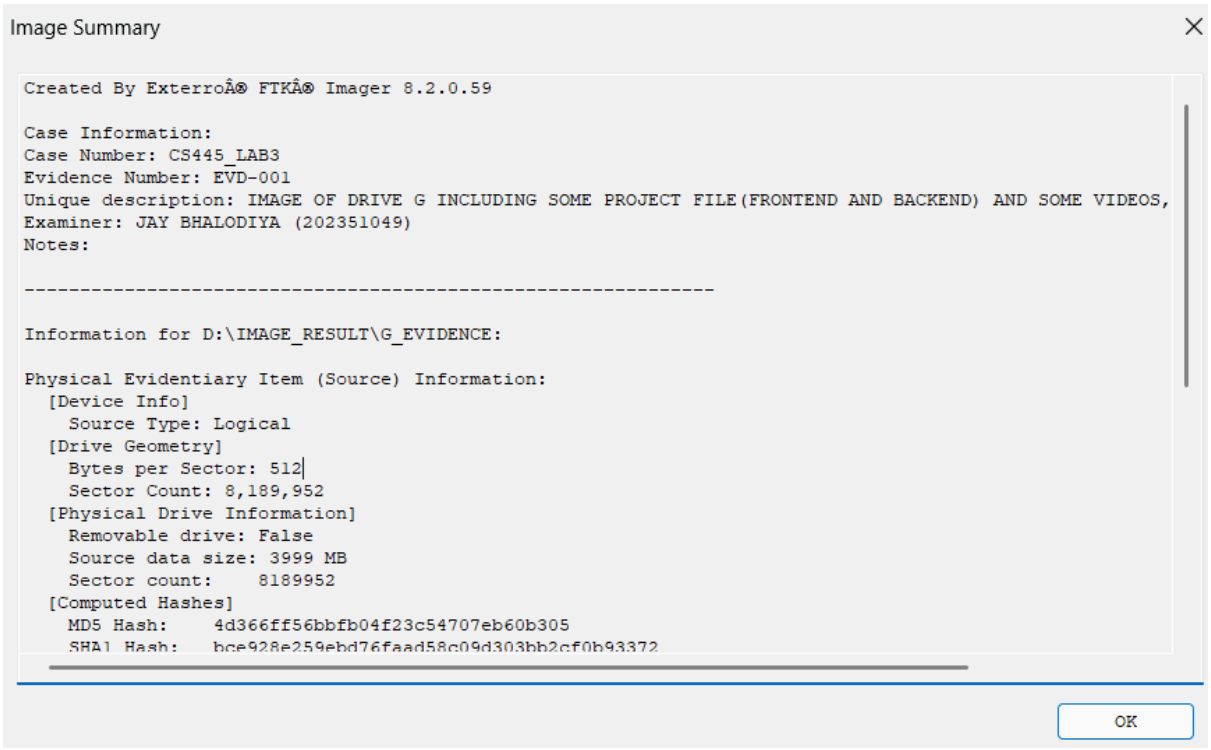
Evidence Number: EVD-001

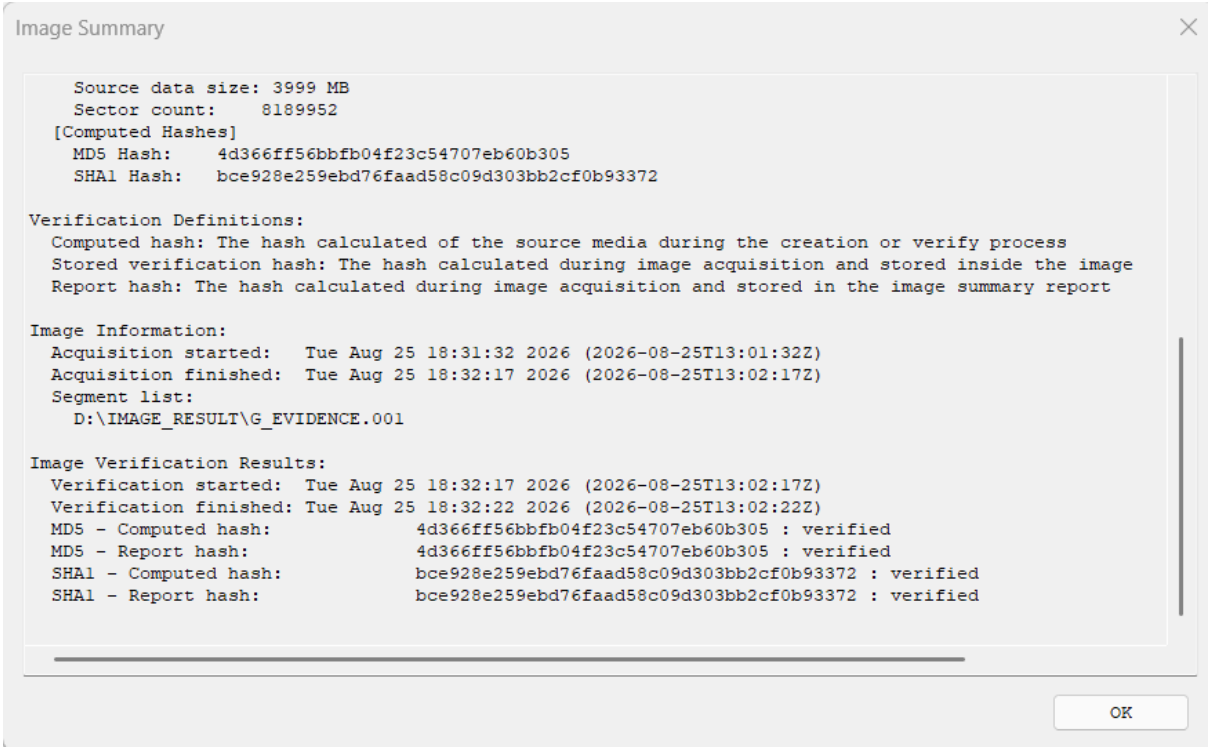
Examiner Name: JAY BHALODIYA (202351049)

Unique Description: Image of Drive G including project files (frontend and backend), videos, picture, and txt files

I selected Drive G as the source drive and set the destination format to Raw (.001), saving it to D:\IMAGE_RESULT\G_EVIDENCE.001. Immediately after image creation, FTK Imager ran an automated hash verification process between the

source drive and the forensic image.





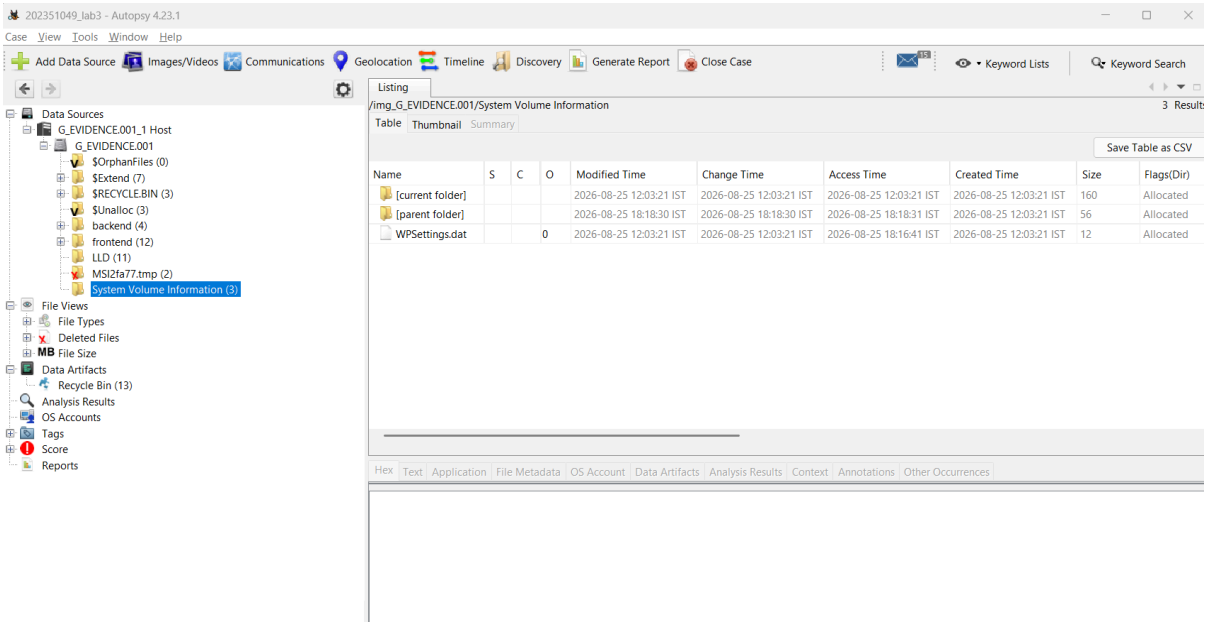
2. File Analysis

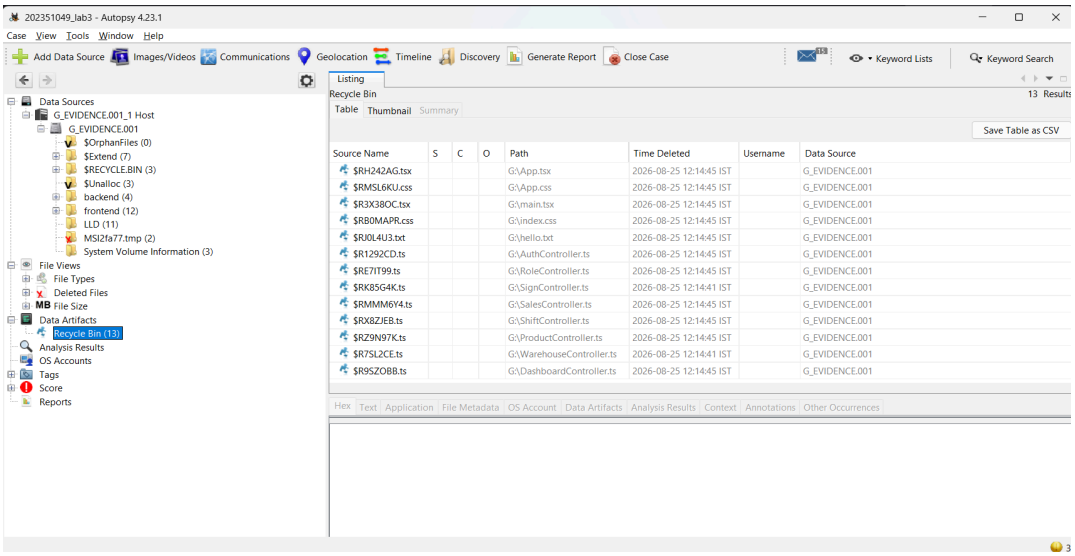
After creating the forensic image, I opened Autopsy to analyze the file system structure. I created a new case named 202351049_lab3 and added G_EVIDENCE.001 as a Disk Image data source and ran the process

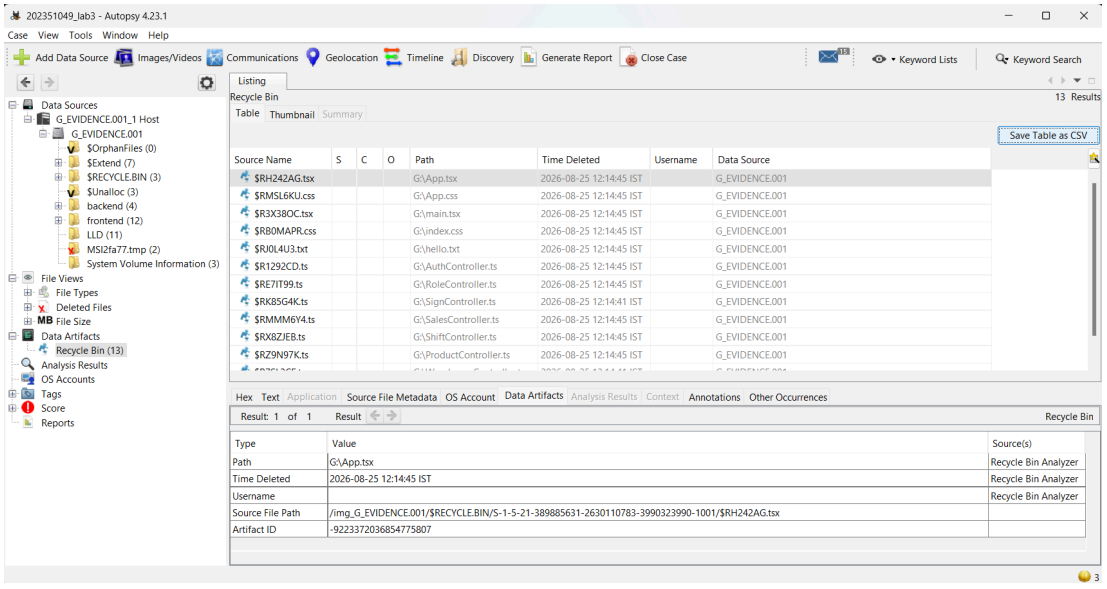
I checked for web browser history under Data Artifacts in Autopsy, but none was found because Drive G was a newly created partition and web browsers store data on C drive. For docx I navigated to File Views > File Types > By Extension > Documents to look for .docx files and no Word documents were found on the drive. The drive only contained project source code files (such as .ts, .tsx, and .css), system design videos, images and plain .txt files.

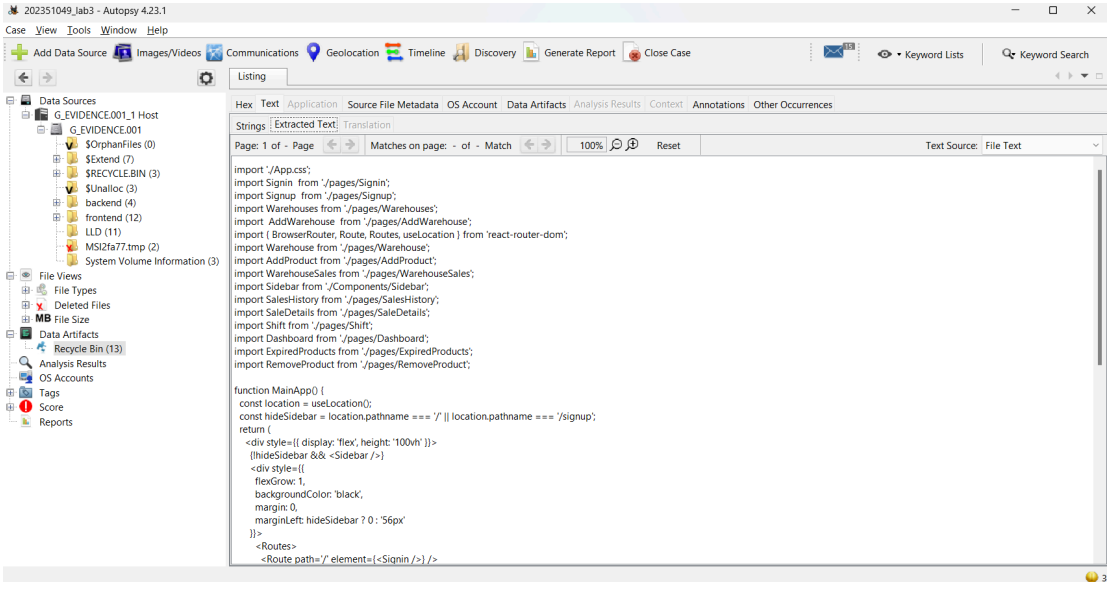
I expanded the recycle bin folder which showed 13 deleted files. I examined a React TypeScript file named App.tsx (originally located at G:\App.tsx and then deleted later) which was deleted on August 25 at 12:14 under the recycle bin path /$RECYCLE.BIN/S-1-5-21-389885631-2630110783-3990323990-1001/$RH242AG.tsx.

Checking the Extracted Text tab in Autopsy confirmed that the full source code was completely readable and uncorrupted showing the whole code. The deleted files included App.css, main.tsx, index.css, hello.txt, AuthController.ts, RoleController.ts, SignController.ts , SalesController.ts, ShiftController.ts, and ProductController.ts because all of these files were fully intact and I right-clicked on them and extracted them and saved few recovered files. The recovered files are uploaded on the drive.




202351049_lab3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Data Sources
G_EVIDENCE.001_1 Host
G_EVIDENCE.001
$OrphanFiles (0)
$Extend (7)
$RECYCLE.BIN (3)
$Unalloc (3)
backend (4)
frontend (12)
LLD (11)
MSI2fa77.tmp (2)
System Volume Information (3)
File Views
File Types
Deleted Files
File Size
Data Artifacts
Recycle Bin (13)
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Recycle Bin
13 Results
Table Thumbnail Summary
Save Table as CSV
Source Name S C O Path Time Deleted Username Data Source
$RH242AG.tsx G:\App.tsx 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RMSL6KU.css G:\App.css 2026-08-25 12:14:45 IST G_EVIDENCE.001
$R3X38OC.tsx G:\main.tsx 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RB0MAPR.css G:\index.css 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RJ0L4U3.txt G:\hello.txt 2026-08-25 12:14:45 IST G_EVIDENCE.001
$R1292CD.ts G:\AuthController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RE7IT99.ts G:\RoleController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RK85G4K.ts G:\SignController.ts 2026-08-25 12:14:41 IST G_EVIDENCE.001
$RMMM6Y4.ts G:\SalesController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RX8ZJEB.ts G:\ShiftController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RZ9N97K.ts G:\ProductController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$R7SL2CE.ts G:\WarehouseController.ts 2026-08-25 12:14:41 IST G_EVIDENCE.001
$R9SZOBB.ts G:\DashboardController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences


202351049_lab3 - Autopsy 4.23.1
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G_EVIDENCE.001_1 Host
G_EVIDENCE.001
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$Extend (7)
$RECYCLE.BIN (3)
$Unalloc (3)
backend (4)
frontend (12)
LLD (11)
MSI2fa77.tmp (2)
System Volume Information (3)
File Views
File Types
Deleted Files
File Size
Data Artifacts
Recycle Bin (13)
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Recycle Bin
13 Results
Table Thumbnail Summary
Save Table as CSV
Source Name S C O Path Time Deleted Username Data Source
$RH242AG.tsx G:\App.tsx 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RMSL6KU.css G:\App.css 2026-08-25 12:14:45 IST G_EVIDENCE.001
$R3X38OC.tsx G:\main.tsx 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RB0MAPR.css G:\index.css 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RJ0L4U3.txt G:\hello.txt 2026-08-25 12:14:45 IST G_EVIDENCE.001
$R1292CD.ts G:\AuthController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RE7IT99.ts G:\RoleController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RK85G4K.ts G:\SignController.ts 2026-08-25 12:14:41 IST G_EVIDENCE.001
$RMMM6Y4.ts G:\SalesController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RX8ZJEB.ts G:\ShiftController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
$RZ9N97K.ts G:\ProductController.ts 2026-08-25 12:14:45 IST G_EVIDENCE.001
Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
Result: 1 of 1 Result
Recycle Bin
Type Value Source(s)
Path G:\App.tsx Recycle Bin Analyzer
Time Deleted 2026-08-25 12:14:45 IST Recycle Bin Analyzer
Username Recycle Bin Analyzer
Source File Path /img_G_EVIDENCE.001/$RECYCLE.BIN/S-1-5-21-389885631-2630110783-3990323990-1001/$RH242AG.tsx
Artifact ID -9223372036854775807


202351049_lab3 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Data Sources
G_EVIDENCE.001_1 Host
G_EVIDENCE.001
$OrphanFiles (0)
$Extend (7)
$RECYCLE.BIN (3)
$Unalloc (3)
backend (4)
frontend (12)
LLD (11)
MSI2fa77.tmp (2)
System Volume Information (3)
File Views
File Types
Deleted Files
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Data Artifacts
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Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
Strings Extracted Text Translation
Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text
import './App.css';
import Signin from './pages/Signin';
import Signup from './pages/Signup';
import Warehouses from './pages/Warehouses';
import AddWarehouse from './pages/AddWarehouse';
import { BrowserRouter, Route, Routes, useLocation } from 'react-router-dom';
import Warehouse from './pages/Warehouse';
import AddProduct from './pages/AddProduct';
import WarehouseSales from './pages/WarehouseSales';
import Sidebar from './Components/Sidebar';
import SalesHistory from './pages/SalesHistory';
import SaleDetails from './pages/SaleDetails';
import Shift from './pages/Shift';
import Dashboard from './pages/Dashboard';
import ExpiredProducts from './pages/ExpiredProducts';
import RemoveProduct from './pages/RemoveProduct';
function MainApp() {
const location = useLocation();
const hideSidebar = location.pathname === '/' || location.pathname === '/signup';
return (
<div style={{ display: 'flex', height: '100vh' }}>
{!hideSidebar && <Sidebar />}
<div style={{
flexGrow: 1,
backgroundColor: 'black',
margin: 0,
marginLeft: hideSidebar ? 0 : '56px'
}}>
<Routes>
<Route path='/' element={<Signin />} />
<Route path='/signup' element={<Signup />} />

```
        <Routes>
          <Route path="/" element={<Signin />} />
          <Route path="/signup" element={<Signup />} />
          <Route path="/warehouses" element={<Warehouses />} />
          <Route path="/addwarehouse" element={<AddWarehouse />} />
          <Route path="/warehouse" element={<Warehouse />} />
          <Route path="/addproduct" element={<AddProduct />} />
          <Route path="/warehousesales" element={<WarehouseSales />} />
          <Route path="/sales/history" element={<SalesHistory />} />
          <Route path="/sales/:id" element={<SaleDetails />} />
          <Route path="/shift" element={<Shift />} />
          <Route path="/dashboard" element={<Dashboard />} />
          <Route path="/expired-products" element={<ExpiredProducts />} />
          <Route path="/removeproduct" element={<RemoveProduct />} />
        </Routes>
      </div>
    </div>
  );
}

function App() {
  return (
    <BrowserRouter>
      <MainApp />
    </BrowserRouter>
  );
}

export default App;

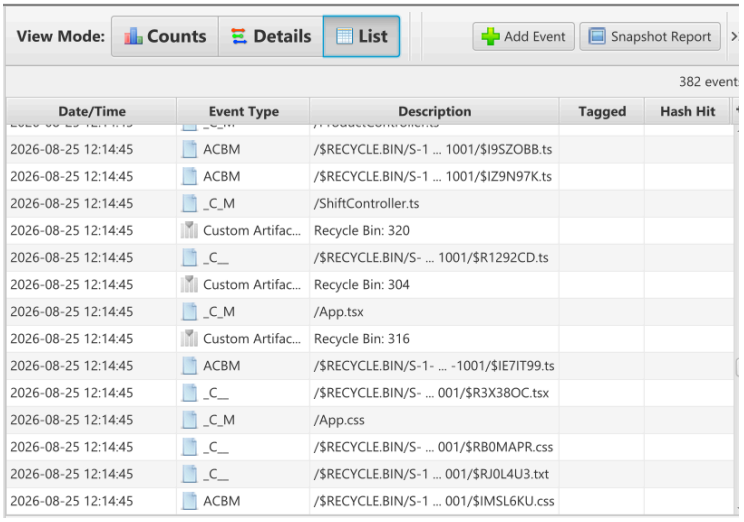
--------------------------------METADATA---------------------------------
```

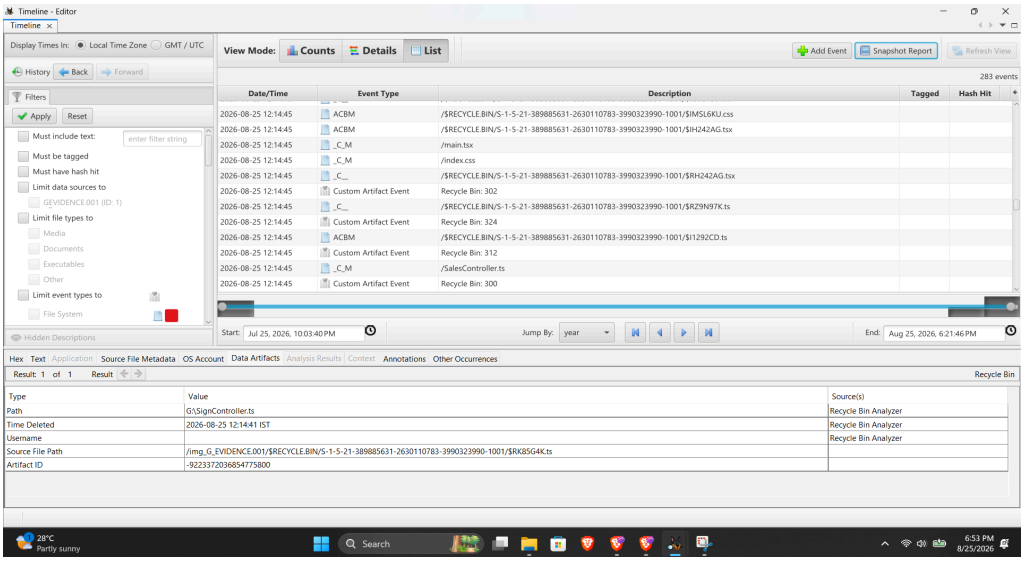
3. Timeline Analysis

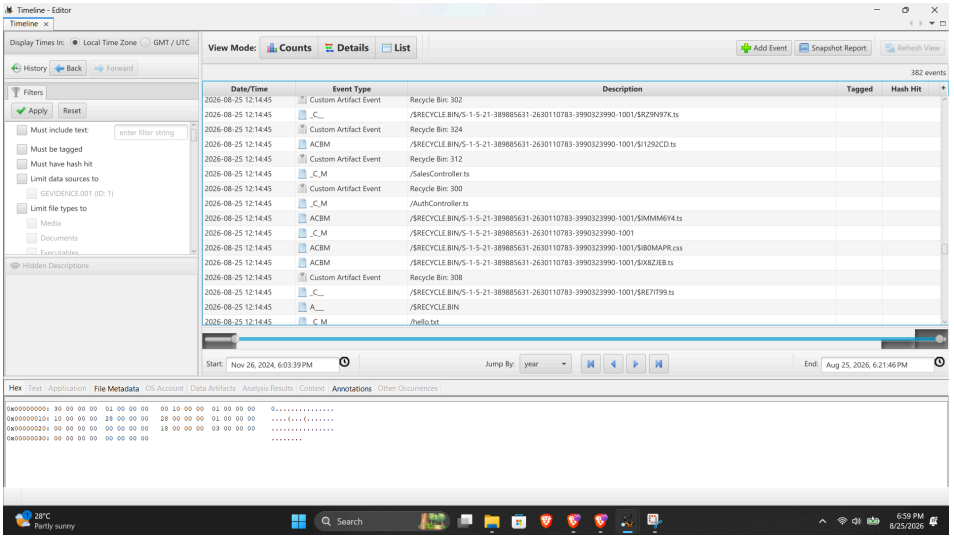
To generate the timeline of events I clicked on the Timeline tool from the top toolbar in the Autopsy interface. I selected the list view to see all file system activities arranged chronologically by their exact timestamps. At 12:11 I was adding the files to the G drive.

2026-08-25 12:11:37	A_B_	/$RECYCLE.BIN/S- ... 001/$R3X38OC.tsx	
2026-08-25 12:11:37	A_B_	/App.css	
2026-08-25 12:11:37	A_B_	/index.css	
2026-08-25 12:11:37	A_B_	/$RECYCLE.BIN/S- ... 001/$RMSL6KU.css	
2026-08-25 12:11:37	A_B_	/App.tsx	
2026-08-25 12:13:23	__B_	/DashboardController.ts	
2026-08-25 12:13:23	__B_	/RoleController.ts	
2026-08-25 12:13:23	__B_	/$RECYCLE.BIN/S-1 ... 1001/$R7SL2CE.ts	
2026-08-25 12:13:23	__B_	/ShiftController.ts	
2026-08-25 12:13:23	__B_	/$RECYCLE.BIN/S- ... 1001/$R1292CD.ts	
2026-08-25 12:13:23	__B_	/ProductController.ts	
2026-08-25 12:13:23	__B_	/$RECYCLE.BIN/S- ... 1001/$RZ9N97K.ts	
2026-08-25 12:13:23	__B_	/SignController.ts	
2026-08-25 12:13:23	__B	/$RECYCLE.BIN/S-1 ... 1001/$RK85G4K.ts	

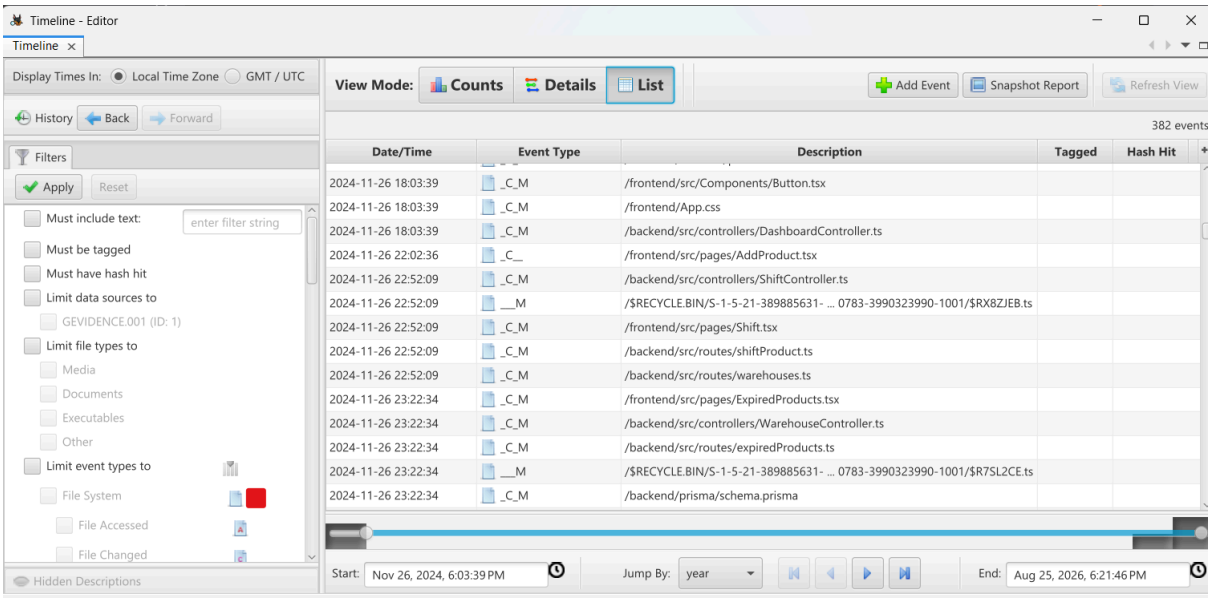
At 12:14 I started deleting the 13 files together.







The timeline showed that after the project files were placed on Drive G starting around 12:03 PM then deletion occurred at 12:14 PM where 13 files were deleted. The timeline also showed when the files were modified as when I was making the project in 2024 and all the history is present.



Recycle Bin Timeline CSV

"Source Name",Path,"Time Deleted",Username,"Data Source"

"$RH242AG.tsx","G:\App.tsx","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RMSL6KU.css","G:\App.css","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$R3X38OC.tsx","G:\main.tsx","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RB0MAPR.css","G:\index.css","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RJ0L4U3.txt","G:\hello.txt","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$R1292CD.ts","G:\AuthController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RE7IT99.ts","G:\RoleController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RK85G4K.ts","G:\SignController.ts","2026-08-25 12:14:41 IST",,G_EVIDENCE.001

"$RMMM6Y4.ts","G:\SalesController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RX8ZJEB.ts","G:\ShiftController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$RZ9N97K.ts","G:\ProductController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001

"$R7SL2CE.ts","G:\WarehouseController.ts","2026-08-25 12:14:41 IST",,G_EVIDENCE.001

"$R9SZOBB.ts","G:\DashboardController.ts","2026-08-25 12:14:45 IST",,G_EVIDENCE.001